

Lecture 21–22: Inequalities and Tail Bounds

1 Tail of a Random Variable

1.1 Definition

The tail of a random variable is defined as:

$$P(X > x)$$

This represents the probability that the random variable takes a value greater than a given threshold.

1.2 Motivation for Studying Tails

From the slides (pages 5–7), the importance of tails is motivated through real-world scenarios:

- Jobs delayed more than 24 hours

- Packets dropped due to full buffer

These correspond to rare but extreme events, captured by tail probabilities.

1.3 Key Idea

- Mean captures average behavior (easy to compute)

- Tail captures rare extreme events (hard to compute)

1.4 Why Are Tails Hard to Compute?

- Requires summing many small probabilities

- Leads to complicated, long summations

This difficulty motivates the need for tail bounds instead of exact computation.

2 Tail Examples

2.1 Example 1: Load Distribution Across Servers

2.1.1 Problem Statement

Suppose jobs are distributed among servers at random.

What is:

$$P(\text{a particular server gets } \geq k \text{ jobs})?$$

2.1.2 Reasoning from Slides

Jobs are assigned randomly

Most servers receive few jobs

Occasionally, one server gets many jobs

This event represents a tail event (rare overload).

2.1.3 Mathematical Formulation

This is a tail probability of a Binomial distribution.

2.2 Example 2: Poisson Arrivals

2.2.1 Problem Statement

Jobs arrive according to a Poisson process with rate λ jobs/hour.

What is:

$$P(\text{at least } k \text{ jobs arrive in the first hour})?$$

2.2.2 Reasoning from Slides

Jobs arrive randomly over time

Most hours have moderate arrivals

Sometimes, there is a burst of arrivals

This represents a tail event (sudden spike in load).

2.2.3 Mathematical Formulation

Again, this is a tail probability involving an infinite sum.

3 Why Tail Bounds?

3.1 Problem with Exact Computation

From page 14:

Binomial tail:

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson tail:

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

These are:

Complicated

Long summations

Difficult to compute exactly

3.2 Key Idea (Page 15)

Instead of computing exact values:

Find an upper bound

Easier to compute

Still provides a guarantee

3.3 Conclusion

Core principle:

$$\text{Exact: } P(X \geq k) = ?$$

$$\text{Bound: } P(X \geq k) \leq \text{something simple}$$

Approximate safely instead of computing exactly.

4 Running Example (Key Idea)

We analyze all bounds using a common example.

4.1 Setup

Flip a fair coin n times

Let:

X = number of heads

4.2 Distribution

$$X \sim \text{Binomial}(n, \frac{1}{2})$$

4.3 Mean

$$E[X] = \frac{n}{2}$$

4.4 Tail Question

$$P\left(X \geq \frac{3n}{4}\right)$$

This represents getting unusually many heads, i.e., a tail region.

5 Markov's Inequality

5.1 Key Idea (Page 18)

Uses only the mean

Large values are rare

Example intuition:

If average is 10, very few values can be 100+

5.2 Statement of Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

5.3 Proof (Step-by-Step from Slides)

Start with expectation:

$$E[X] = \sum_{x=0}^{\infty} x p_X(x)$$

Step 1: Restrict to large values

$$E[X] \geq \sum_{x=a}^{\infty} x p_X(x)$$

Step 2: Replace by a (since $x \geq a$)

$$\geq \sum_{x=a}^{\infty} a p_X(x)$$

Step 3: Factor out a

$$= a \sum_{x=a}^{\infty} p_X(x)$$

Step 4: Recognize probability

$$= a \cdot P(X \geq a)$$

Final inequality:

$$\begin{aligned} E[X] &\geq a \cdot P(X \geq a) \\ \Rightarrow P(X \geq a) &\leq \frac{E[X]}{a} \end{aligned}$$

6 Markov's Inequality on Running Example

Given:

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad E[X] = \frac{n}{2}$$

Apply Markov's Inequality:

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{E[X]}{3n/4}$$

Substitute mean:

$$\begin{aligned} &= \frac{n/2}{3n/4} \\ &= \frac{2}{3} \end{aligned}$$

6.1 Conclusion

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{2}{3}$$

6.2 Observation (from Slide)

This is a terrible bound

Reason: it does not depend on n

7 Real-Life Interpretation

7.1 Example: Income Distribution

Average income = 10,000

Using Markov's inequality:

Very high incomes must be rare, since the average is limited

This demonstrates how mean alone can bound extreme outcomes.

8 Case Study Motivation

From the final slide:

How to predict the worst-case demands on a server?

This directly connects to:

Tail probabilities

Bounding extreme events

Designing systems robust to rare spikes

9 Final Summary

1. Tail probability measures extreme events:

$$P(X > x)$$

2. Exact computation is difficult due to long summations

3. Tail bounds provide simpler, safe approximations

4. Markov's inequality:

Uses only the mean

Provides a simple upper bound

Often loose (weak)

5. Running example shows:

Tail events = rare

Bounds may lack tightness